DENGUEWATCH: AN AI-ASSISTED PREDICTIVE DENGUE SURVEILLANCE AND HOTSPOT ANALYTICS WITH GEOGRAPHIC INFORMATION SYSTEM

An Undergraduate Capstone Project Presented to the
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Chapter 1

PROBLEM AND ITS SETTING

Introduction

Dengue is mainly spread by Aedes aegypti mosquitoes, which breed more faster during the rainy season, leading to millions of infections each year. According to the World Health Organization (WHO), dengue cases are increasing because of climate change, rapid urbanization, and poor sanitation. Weather conditions, including rainfall, temperature, and humidity, also influence mosquito breeding and the spread of dengue. Therefore, incorporating weather data into dengue prediction systems can help identify hotspot areas and provide earlier warnings [1].

In the Philippines, dengue continues to be a serious public health problem. According to the Department of Health (DOH), in 2024, there was a 15% increase in the total dengue cases in the country with 77, 867 cases and 205 deaths from January 1 through January 15, 2024, compared to 64,905 cases and 171 deaths during the same period last year. A total of 62,313 were reported in the first 3 months of 2025, up to 73% from 338,983 cases during the same time frame in 2024. The continuous increase in dengue cases is a clear indication of the reason why Dengue remains a public health concern to this day and the continuous efforts in both dengue prevention and control efforts. The fact that the cases are not timely detected, followed up and monitored in varied regions,

is a major issue. Reports of dengue cases are delayed, and health data is manually collected and processed leading to incomplete data and delays. The difficulties of these challenges make it challenging to promptly identify the location of dengue hotspots, and act faster in responding to the outbreak. This has led to the need of a data-driven system for early prediction; continuous surveillance and monitoring; and immediate identification of high risk areas for effective country-wide prevention and control of dengue [2] [3] [4].

Dengue is still one of the major public health problems in Iligan City, especially during the rainy and flood season when mosquitoes reproduce rapidly. Poor sanitation and improper waste management in some areas also contribute to the spread of mosquito breeding sites. The seriousness of the problem was highlighted when the city government declared a state of calamity after recording 1,138 dengue cases and 14 dengue-related deaths from January 1 to August 17 year 2019, with the highest number of dengue deaths in Northern Mindanao during that period. To respond to the outbreak, the local government allocated more than PHP 8 million from its quick response fund for protective equipment, fogging machines, medicines, mosquito nets, chemicals, and public awareness campaigns. The City Health Office of Iligan also reported that the city recorded dengue cases in 2024, with a similar number of cases reported in 2025. The city remains concerned about the increasing number of dengue cases, particularly in densely populated and flood-prone barangays [5].

To solve these problem, a data-driven dengue monitoring and prediction system can aid the health officials and the citizens of Iligan City to better track these dengue cases. The system can analyze case data, maps and predictive analytics to generate hotspot locations, trends and early warning of prospective epidemics. The technology can also be used to build the capacity of LGUs and city health workers to respond swiftly, improving dengue prevention and decision making faster.

There are a lot of innovations in the field but many local health systems continue to struggle to use these tools to their full potential, especially due to resource limitations and the lack of integrated data systems and also limited technical capacity to analyze and make decisions in real-time [6].

Geographic Information System (GIS) is an important technology used in public health to collect, manage, analyze, and display location-based data. In dengue surveillance, GIS helps health authorities map dengue cases, identify hotspot areas, and monitor where the disease is spreading. It also helps show which communities are at higher risk by displaying dengue cases on digital maps. When combined with predictive analytics and artificial intelligence, GIS can help detect possible outbreaks earlier and support faster decision-making. This allows local governments and health workers to focus their prevention efforts on high-risk areas, improve resource allocation, and respond more quickly to dengue outbreaks. Because of these benefits, GIS has become an effective tool for improving dengue surveillance and disease prevention [1][7].

Statement of the Problem

Dengue monitoring and prevention in Iligan City continue to face numerous problems, affecting quick responses to outbreaks. These challenges include:

1. The manual collection of dengue data causes delays in reporting and monitoring, which makes it challenging for City Health Workers to keep updated on dengue in different barangays and demands a significant investment of time and energy.
2. It is challenging for city health workers and residents to recognize which areas are most affected by dengue because there are few tools available to accurately identify and display areas with high numbers of dengue cases.
3. Current dengue surveillance procedures cannot forecast new dengue hotspot locations because these procedures only use historical records of dengue cases and do not consider weather data such as rainfall, temperature, and humidity, which influence mosquito breeding and dengue transmission.
4. During dengue outbreaks, coordination and information sharing between Barangay Health Workers, City Health Workers, and Local Government Units can be inefficient and slow, delaying the sending of alerts, warnings, and information to affected communities.

Objective of the Study

The system will help monitor dengue cases, predict possible hotspot areas, display dengue data on maps, and support faster response to possible dengue outbreaks. These are the following specific objectives:

1. To design a system can digitally gather, organize, and monitor dengue-related data in order to minimize reporting delays and save City Health Workers' time and effort.
2. To develop and evaluate an artificial intelligence (AI)-based predictive model that integrates historical dengue case data, geographic information system (GIS) data, and weather data to predict potential future dengue hotspots.
3. To evaluate a fast and accurate prediction system that can locate possible dengue hotspot locations in Iligan City, allowing for timely response and preventive measures against dengue outbreaks.
4. To test the system's effectiveness in increasing awareness and help in preventing the outbreak of dengue outbreaks by giving locals timely dengue alerts and risk information.

Scope and Limitation

This capstone project focuses on the development of DengueWatch: An AI-Asssited Predictive Dengue Surveillance and Hotspot Analytics with Geographic Information System, a web and mobile-based system designed to improve dengue monitoring, hotspot prediction, data management, and information sharing in Iligan City. The system predicts possible dengue hotspot areas by analyzing historical dengue case records, current dengue case reports, GIS location data, and weather information such as rainfall, temperature, and humidity using the Random Forest algorithm. It provides features including dengue case reporting, GIS-based hotspot mapping, trend visualization, risk level classification, alerts and notifications, report generation, and user management.

The system will be used by the City Health Office (CHO), City Health Workers (CHWs), Barangay Health Workers (BHWs), Local Government Units (LGUs), and residents through web and mobile applications. The City Health Office will manage the system and oversee dengue monitoring activities. City Health Workers will record and manage dengue case reports throughout the city, while Barangay Health Workers will monitor and report dengue cases within their respective barangays. Local Government Units can use the generated reports and hotspot maps to support planning and dengue prevention activities, while residents can receive dengue alerts, preventive reminders, and hotspot updates. Thus, this system is expected to raise community awareness, enhance

dengue surveillance in the city, and enable a rapid response to dengue outbreaks in Iligan City.

However, this study has several limitations. The system requires a stable internet connection for real-time reporting, synchronization, and access to web and mobile services. The prediction of dengue hotspots depends on the quality, completeness, and timeliness of the historical dengue records and the current dengue case reports, geographic information, and weather data used by the model. The use of incomplete, inaccurate, or delayed data in any of these may adversely affect the prediction. Finally, this system only focuses on dengue surveillance and hotspot prediction within Iligan City and does not consider other diseases or locations outside Iligan City. Most crucially, as the system's predictions are only meant to assist with monitoring and decision-making, it should not be seen as a completely accurate substitute for medical or epidemiological assessment.

Significance of the Study

The following entities that will benefit from this study are:

For the City Health Office (CHO). The study provides a monitoring dengue case trends, identifying the hotspot locations, and managing dengue records by facilitating faster reporting, improve data management, and for better decision-making of a dengue prevention.

For City Health Workers. Health professionals can respond to possible outbreaks more quickly due to the system's access to automated reporting, citywide case tracking, and current dengue information.

For Barangay Health Workers. Facilitates faster submission of dengue cases within each barangays through better coordination and information exchange with the City Health Office.

For Local Government Units (LGUs). LGUs may identify high-risk areas and organize dengue prevention measures including fogging operations, clean-up drives, and awareness campaigns, and resource allocation with the use of the generated hotspot maps, alerts, and reports.

For Community Residents. The system may increases public awareness and encourages local participation in dengue prevention activities by providing a timely dengue alerts, hotspot updates, and prevention reminders.

For the Public Health Sector. By providing a tool that allows early detection of dengue hotspot locations, enhances monitoring efforts, and helps health officials

make evidence-based decisions for dengue prevention and response, the study supports data-driven public health management.

For Present Researchers and Future Developers. Future researchers and the Future Developers solutions for dengue monitoring, prediction, and public health management may use this study as a guide.

Definition of Terms

Aedes aegypti. A type of mosquito that mainly spreads the dengue virus. It is the primary carrier of dengue in the community [8].

Alert Notification. Automated messages sent to users regarding dengue risks [9].

Cloud-Based Database. An online database used to store and manage dengue - related information that can be accessed through the internet 10].

Data-Driven System. A system that uses collected data to monitor dengue cases, predict hotspots, and support decision-making[11].

Dengue. A viral disease spread by Aedes aegypti mosquitoes. It is the disease being monitored in this study 12].

Dengue Monitoring. The process of collecting and tracking dengue case information to identify trends and affected areas[13].

DengueWatch. The proposed system developed in the study [14].

Geotagging. The process of adding location information to dengue case records for mapping and hotspots identification [15].

Health Workers. Nurses, barangay health workers, and public health staff who manage and report dengue-related information in the system [16].

Hotspot Area. A place identified as having a high number of dengue cases or a high risk of outbreaks [17].

Hotspot Mapping. Visualization of high-risk dengue areas on a digital map [18].

Local Government Units (LGUs). Government offices responsible for public health programs and dengue prevention in Iligan City [19].

Machine Learning. A technology that learns from data to make predictions. In this study, it is used to predict dengue hotspot areas [20].

Mobile Application. The mobile version of the DengueWatch system that allows users to access dengue information, alerts, and monitoring features through smartphones or tables [21].

Prediction Model. The Algorithm can be used this to forecast dengue hotspot areas [22].

Predictive. The ability of the system to forecast a potential dengue hotspot areas using the available dat**a** [23].

Web-Based System. The web version of DengueWatch allows authorized users to led and monitor the dengue information through a web browser [24].

Chapter 2

RELATED LITERATURE AND RELATED STUDIES

This chapter presents the related literature and related studies that support the development of the proposed DengueWatch: An AI-Assisted Predictive Dengue Surveillance and Hotspot Analytics with Geographic Information System. The materials reviewed in this chapter establish the machine learning, artificial intelligence and epidemiological foundation for the system's design and identify the specific gaps that the proposed system addresses.

Related Literature

Philippines Dengue Cases on the Rise Dengue cases in the Philippines were highlighted, stressing the need for more careful surveillance and preventive measures. The Philippine News Agency story said local government entities and health agencies should improve their surveillance systems to detect areas where dengue prevalence is increasing and respond to communities hit by the disease. The research also stressed that dengue is a huge public health concern in the entire country, not just in Iligan City. This situation underscores the need of monitoring and forecasting dengue epidemics by technological and data-driven approaches [25].

Dengue remains a major public health concern in the Philippines because the country's tropical climate, heavy rainfall, rapid urbanization, and inadequate

waste management create favorable conditions for mosquito breeding throughout the year. The Department of Health continues to report thousands of dengue cases annually, making dengue one of the country's most common vector-borne diseases. In Iligan City, dengue is particularly significant because the city has experienced recurring outbreaks, including the declaration of a state of calamity after recording 1,138 cases and 14 deaths in a single outbreak period. Flood-prone barangays, high population density, and environmental conditions increase the city's vulnerability to dengue transmission. These challenges highlight the need for an AI-assisted predictive dengue surveillance and hotspot analytics system that enables health authorities to detect high-risk areas early, improve decision-making, and implement timely interventions before outbreaks worsen [2][3][4][5].

2.1 **Dengue Surveillance System**

Dengue surveillance systems play an important role in monitoring, collecting, analyzing, and reporting dengue cases to support public health decision-making. These systems enable health authorities to identify disease trends, monitor outbreak patterns, detect high-risk communities, and evaluate the effectiveness of dengue prevention programs. Conventional surveillance systems often rely on manual reporting from hospitals, rural health units, and local government offices. While these methods remain useful, they frequently experience delays in data submission, processing, and analysis, making it difficult for health officials to respond immediately during rapidly increasing

outbreaks. Recent studies have shown that digital surveillance systems significantly improve the efficiency of disease monitoring by providing centralized databases, automated data processing, and real-time reporting. Modern surveillance platforms also integrate mapping technologies, predictive models, and data visualization tools that enable health authorities to identify dengue hotspots and allocate resources more effectively. As public health continues to adopt digital transformation, integrating intelligent surveillance systems into dengue monitoring has become an important strategy for improving outbreak preparedness and supporting evidence-based decision-making [26].

2.2 Geographic Information System

Geographic Information System (GIS) has become one of the most valuable technologies used in disease surveillance and public health management. GIS allows health authorities to collect, manage, analyze, and visualize geographically referenced health data, making it easier to identify the spatial distribution of diseases such as dengue. By displaying dengue cases on interactive digital maps, GIS enables decision-makers to detect hotspot locations, monitor disease spread, and identify environmental factors that contribute to mosquito breeding. Spatial analysis provided by GIS also assists health workers in determining which communities require immediate intervention, allowing resources to be distributed more efficiently. Several studies have demonstrated that GIS improves disease surveillance by providing visual representations of

outbreak patterns that cannot easily be identified through traditional statistical reports alone. When combined with historical dengue records, climate information, and demographic data, GIS becomes an effective decision-support tool for predicting future outbreaks and strengthening public health planning. Because of these capabilities, GIS has become an essential component of many modern dengue surveillance systems worldwide [27].

2.3 Artificial Intelligence and Machine Learning for Dengue Prediction

Artificial Intelligence (AI) and Machine Learning (ML) have gained significant attention in recent years because of their ability to analyze large volumes of health data and generate accurate disease predictions. Unlike traditional statistical methods, machine learning algorithms can identify complex relationships among historical dengue cases, climate variables, rainfall, temperature, humidity, vegetation, and environmental conditions that influence mosquito populations. Various machine learning techniques, including Random Forest, Decision Tree, Support Vector Machine, Artificial Neural Network, and deep learning models, have demonstrated high prediction accuracy in forecasting dengue outbreaks. These technologies enable health authorities to estimate future disease incidence several weeks before outbreaks occur, allowing preventive actions to be implemented earlier. AI-powered prediction models also continuously improve their performance as additional data become available, making them highly suitable for long-term disease surveillance. The integration

of artificial intelligence into dengue monitoring systems enhances the ability of local governments to identify high-risk communities, prioritize interventions, and optimize the allocation of healthcare resources [28].

2.4 Predictive Analytics in Public Health

Predictive analytics refers to the process of analyzing historical and current data using statistical methods, data mining, and machine learning techniques to forecast future events. In public health, predictive analytics has become an important tool for anticipating disease outbreaks, estimating healthcare demands, and supporting evidence-based planning. For dengue surveillance, predictive models analyze historical case records together with environmental, meteorological, and demographic data to identify communities that are likely to experience future outbreaks. This information allows health authorities to implement mosquito control activities, community awareness campaigns, and medical resource preparation before disease transmission increases significantly. Several studies have shown that predictive analytics improves public health decision-making by providing early warnings and reducing response time during disease outbreaks. As governments continue adopting digital technologies in healthcare, predictive analytics has become an important component of intelligent disease surveillance systems capable of supporting proactive rather than reactive public health interventions [29].

2.5 Early Warning System for Dengue

Early warning systems (EWS) for dengue are designed to detect and predict potential dengue outbreaks before they become widespread. These systems collect and analyze data from multiple sources, including historical dengue cases, climate conditions such as rainfall, temperature, and humidity, environmental factors, mosquito surveillance, and population information. By using predictive models and advanced data analysis techniques, early warning systems can estimate the likelihood of future dengue outbreaks and identify communities that are at greater risk. This allows health authorities to receive timely information that supports faster and more effective decision-making [30].

Related Studies

According to the study by Modeling Dengue Cases and Online Search Behavior for Prediction Models mentioned that dengue is still a huge public health problem because it still affects so many people. Health organizations are finding it difficult to trace the cases of dengue, their locations and the intensity of epidemics in local communities, the research said. It also shows the potential of technologies such as social media and internet searches to complement dengue surveillance by raising awareness and facilitating the dissemination of information on dengue-related activities. The study employs several statistical techniques to forecast possible dengue outbreaks. The results suggest that in some of the selected cities the dengue cases continue to increase over time. The accuracy of the forecasting models was assessed using measures like mean

absolute error, mean squared error, and root mean-square deviation. Moreover, the study indicated that monitoring and predicting dengue transmission in many areas might be enhanced by using additional online data sources, such as social media [31].

A study by Acharjee et al. (2022) proposed a machine learning approach to predict dengue fever using patient medical records, symptoms, and diagnosis reports collected from hospitals in Bangladesh. The researchers developed a dataset from real dengue patients and applied two machine learning algorithms, namely Decision Tree (DT) and Random Forest (RF), to classify dengue cases. The results showed that the Decision Tree algorithm achieved an accuracy of 79%, which performed better than the Random Forest model. The study concluded that machine learning can help predict dengue cases at an early stage, allowing healthcare professionals to provide timely treatment and reduce the impact of dengue outbreaks. The findings support the use of artificial intelligence in disease prediction and demonstrate its potential to improve dengue surveillance and public health decision-making. This study is related to the proposed denguewatch system because both utilize machine learning techniques to improve dengue prediction and assist health authorities in implementing early interventions. [32].

DengueGNN: Graph-based deep learning for modeling disease spread dynamics and prediction, a similar paper, proposes to employ a Dynamic Spatio-Temporal Graph Neural Network (ST-GNN) model to enhance regional

dengue prediction. Dengue is a serious public health problem, the research says, because of factors such as human mobility, climate change and regional links that enable it to spread globally at a rapid pace. However, typical statistical and machine learning models have been found to struggle to capture the complex spatial and temporal patterns of dengue transmission, which can result in poor forecasts and a delayed response to epidemics . To boost prediction accuracy, the researchers constructed a spatiotemporal graph neural network (ST-GNN) model that incorporates time and space data. Regions were represented as graph nodes and human movement and nearby regional linkages as edges in the model. The system also included past dengue case data and environmental characteristics. The study showed that the ST-GNN model performed better than ten baseline models [33].

Using mosquito-based surveillance to improve dengue monitoring and outbreaks identification was examined in Monitoring Dengue Virus in Aedes aegypti to Improve Dengue Surveillance and Control in Puerto Rico. The study says that standard dengue prediction often depends on the reported human cases, which may not totally reflect transmission because some infections are undetectable. To address this problem, the researchers monitored Aedes aegypti mosquitoes in 15 Puerto Rican areas or locations during the year 2024 dengue outbreak. Every week, adult female mosquitoes were gathered and subjected to RT-PCR testing for the dengue virus (DENV).Different dengue were found in a number of localities, according to the results, and mosquito-based surveillance

revealed more virus activity than passive human case reporting. According to the study, integrating human and mosquito surveillance can enhance outbreak detection, bolster vector management tactics, and lessen dengue transmission in impacted areas. [34].

Neural networks for dengue forecasting: a systematic evaluation of more than sixty scientific publications looked at the application of neural network models in predicting dengue and other illnesses. This study shows that data that are collected at the city, municipality or barangay level are more accurate in prediction conclusions than national-level information. It also highlighted the importance of including web-based dashboards and visualization tools to help public health workers in monitoring sickness trends and understanding prediction results. Results suggest that predictive analytics, along with widely available monitoring platforms, can strengthen dengue surveillance, early warning systems and outbreak response [35].

Table 2
Research Synthesis Matrix

Related Studies	Specific Objectives	Methodology	Problem Unknown/Gap	Remarks
J. J. M. Ecleo, A. P. Galido (2024)	The study aims to predict dengue outbreaks by looking at what people search for on Google (using Google Trends) and comparing it to actual dengue cases to help create early warning systems.	The researchers used a math-based approach to see how online search habits match real dengue records over time. They tested different prediction models (like ARIMA and machine learning) to find the most accurate one.	Traditional health reporting takes too long to gather and share data. While internet searches can give quick hints about outbreaks, it is still hard to know if online searches always match real clinical cases accurately.	While this study uses internet search data to find trends, or send direct alerts to the public. Our system, DengueWatch, improves on this by using actual local case data to predict risks. We also build active web and mobile tools with mapping and push notifications for health workers and residents using Django, HTML, CSS, and Python.

Table 2

Table 2
Research Synthesis Matrix (Continuation)

Related Studies	Specific Objectives	Methodology	Problem Unknown/ Gap	Remarks
Sarma, D., Hossain, S., Mittra, T., Bhuiya, M. A. M., Saha, I., & Chakma, R. (2020).	The study aimed to develop a machine learning model that can predict dengue fever using patients' diagnosis reports, medical history, and symptoms. The objective was to assist healthcare professionals in identifying dengue cases early and providing timely treatment.	The researchers collected real-time patient data from Chittagong Medical College Hospital and Dhaka Medical College Hospital in Bangladesh. The dataset included patients' diagnosis reports, medical history, and symptoms. The data were divided into 70% for training and 30% for testing. Two machine learning algorithms, Decision Tree (DT) and Random Forest (RF), were applied to classify dengue cases. The Decision Tree algorithm achieved the highest accuracy of 79%.	The study focused on predicting dengue using clinical patient data only. It did not include environmental factors, weather information, geographic location, or hotspot mapping, which are important for predicting community-level dengue outbreaks and supporting public health surveillance	This study is related to the proposed DengueWatch because both apply machine learning techniques to improve dengue prediction. However, the proposed system extends the study by integrating Artificial Intelligence, Geographic Information System (GIS), historical dengue records, environmental data, hotspot visualization, and predictive analytics. It also provides an interactive web-based dashboard and real-time monitoring to help local health authorities identify high-risk areas and implement early dengue prevention measures.

Table 2
Table 2
Research Synthesis Matrix (Continuation)

Related Studies	Specific Objectives	Methodology	Problem Unknown/Gap	Remarks
R. B. GulMohamed, W. Hetany, H. A. Almaimani, F. A. S. Khiery (2026)	The study aims to accurately predict how dengue spreads across different regions over time by connecting environmental factors and human movement patterns.	The researchers developed a deep learning model called DengueGNN (Spatio-Temporal Graph Neural Network). It maps out different regions as points on a graph, connects them based on how people travel between them, and uses LSTM networks with an attention mechanism to analyze historical climate and case trends.	Current forecasting models often look at time changes or location changes separately, failing to combine how dynamic human movement networks, geographic proximity, and shifting weather patterns interact to drive regional outbreaks simultaneously.	This study presents an advanced, high-level deep learning approach to track complex disease movement across multi-country datasets. Our system, DengueWatch, takes the core concept of connecting location, environment, and time, but scales it down into a highly practical tool for local deployment. Instead of relying on heavy graph neural network frameworks, we build a streamlined web and mobile platform using Django, HTML, CSS, and Python that takes patterns and case data to create immediate, user-friendly hotspot maps and send direct push notifications to health workers and residents.

Table 2

Table 2
Research Synthesis Matrix (Continuation)

Related Studies	Specific Objectives	Methodology	Problem Unknown/Gap	Remarks
L. M. Otero, et al. (2025)	The study aims to test if tracking the dengue virus directly inside caught mosquitoes (entomo-virological surveillance) can help find local outbreaks early and guide mosquito control efforts better than traditional human health reporting.	Researchers set up special traps (AGO traps) weekly around areas with recent dengue cases. They grouped the trapped female mosquitoes into batches and used laboratory genetic testing (RT-PCR) to look for the dengue virus and its specific variations (serotypes).	Traditional surveillance relies on sick people visiting clinics and doctors reporting them. This method takes too much time and misses a massive number of infections since many people with dengue do not show severe symptoms, leaving local health teams unaware of active mosquito transmission.	This study highlights an advanced biological approach to tracking dengue but requires intensive laboratory testing of actual insects. Our system, DengueWatch, bridges this reporting gap by using easily accessible, field-ready data. Instead of trapping and testing mosquitoes in a lab, we integrate immediate digital inputs—like local case data to instantly display visual hotspot risk maps. Our platform manages this data through role-based web and mobile dashboards that send live push notifications using Django, HTML, CSS, and Python.

Table 2

Table 2
Research Synthesis Matrix (Continuation)

Related Studies	Specific Objectives	Methodology	Problem Unknown/Gap	Remarks
L. Lober, F. A. Rodrigues, K. O. Roster (2025)	The study aims to review how artificial neural networks are used to forecast dengue cases in humans worldwide, helping future researchers design better and more accurate prediction models.	Following strict PRISMA guidelines, the researchers analyzed 62 published papers. They evaluated the technical choices made in those studies, including the types of neural networks used, input data (like climate and population facts), and how well they performed compared to other machine learning models.	While many papers use neural networks for dengue forecasting, there is no unified standard for selecting models, tuning settings, or choosing input features. Additionally, the use of advanced deep learning networks remains largely unexplored, making it hard to find the most reliable method for different areas.	This study provides a wide overview of global predictive models, showing that neural networks perform very well when using city-level data and weekly timelines. Our system, DengueWatch, applies these exact findings by building a localized, real-world platform. Instead of just reviewing models, we put them into practice—combining local weekly cases and environmental factors to generate interactive risk maps. We deliver these tools directly to health workers and residents via web dashboards and mobile alerts using Django, HTML, CSS, and Python.

Table 2

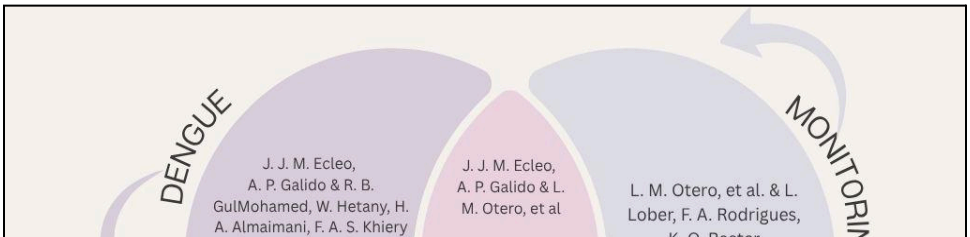


Figure 1. Summary of Related Studies

In conclusion, the related studies show that dengue surveillance, monitoring, and machine learning are important in helping prevent and control dengue. They also show that using AI and Geographic Information System (GIS) can improve the prediction of dengue hotspots and support faster decision-making. However, many previous studies only focused on some of these features. The proposed DengueWatch combines dengue prediction, hotspot monitoring, and GIS mapping in one system. This provides a more complete and useful tool to help health officials monitor dengue cases, identify high-risk areas, and take early action to reduce the spread of the disease.

Chapter 3

RESEARCH METHODS

This chapter presents the methodology used in developing DengueWatch: An AI-Assisted Predictive Dengue Surveillance and Hotspot Analytics with Geographic Information System. The process began with requirements gathering and problem identification, followed by the design of the system based on the findings, and the subsequent development of the analysis models, system design, and training procedure used to build the platform.

3.1 Requirements Gathering and Problem Identification

Before building the system, the researchers first gathered information about how dengue cases are currently being handled in Iligan City. This was done by talking directly with the City Health Office (CHO), City Health Workers (CHWs) through interviews. These conversations helped the researchers understand the real problems these workers face every day. Reports on dengue cases take too long because everything is still done manually, and there is no easy way for workers to see which areas have the most cases. Existing methods also cannot predict where future outbreaks might happen, and sharing information between barangays, city workers, and the local government is often slow and disorganized.

Aside from the interviews, the researchers also looked at past dengue records from the CHO and studied related research about dengue tracking

systems, GIS mapping, and machine learning to help decide what the system needed. Location data for the different barangays and puroks was also collected to support the mapping feature. Based on everything gathered, the researchers decided that DengueWatch should be able to collect dengue reports digitally, show hotspot maps, predict outbreaks using AI, send alerts, and generate reports to help with decision-making.

3.2 Design of the DengueWatch System

After identifying the problems, the researchers designed DengueWatch as a system that uses AI and GIS, or digital maps, to help predict and monitor dengue outbreaks in Iligan City. The goal was to replace the slow, manual, and paper-based process with one central web and mobile system that everyone involved can use.

The system combines three types of information: past dengue case records, newly submitted dengue case reports, and location data taken from Google Maps. All of this is cleaned and organized into one dataset before it is studied by a machine learning model called Random Forest, which was chosen because it works well even with complicated data and gives reliable predictions. Once the model predicts which areas are at risk, the results are shown on a digital map using GIS, with each area labeled according to its risk level as Low, Moderate, High or Critical.

The system was also designed so that different users can access features based on their role. The City Health Office, City Health Workers, Barangay Health Workers, Local Government Units, and Residents each have their own functions, such as reporting cases, viewing hotspot maps, receiving alerts, or generating reports. Through this design, DengueWatch does not only predict dengue hotspots, it also helps everyone respond faster and make better decisions to prevent outbreaks in Iligan City.

Analysis Modeling

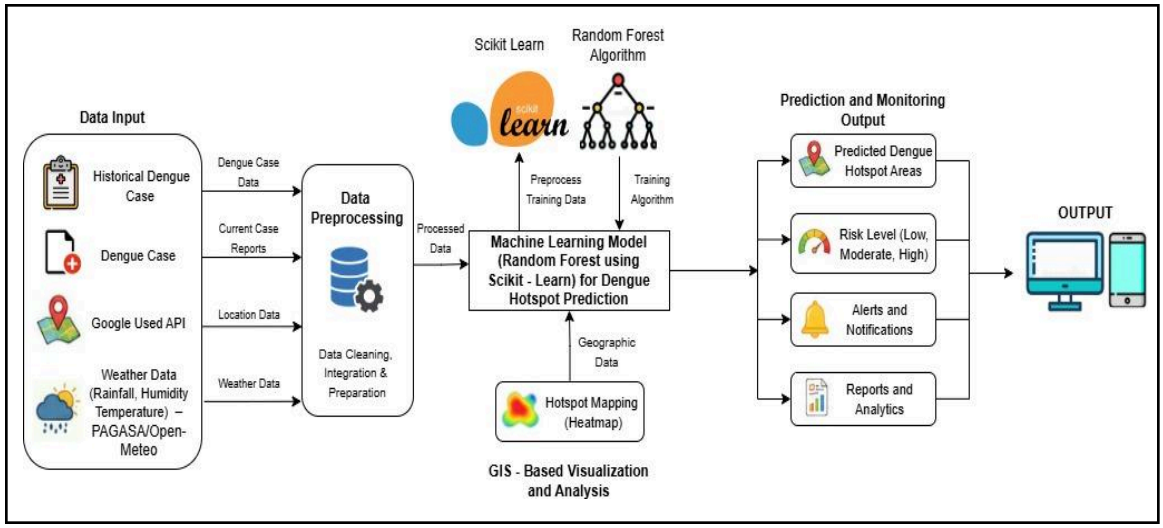


Figure 3.3 System Architecture Design

Figure 3.3 shows the overall workflow of the DengueWatch system for predicting and monitoring dengue hotspot areas in Iligan City. It starts with collecting data from four major sources: previous dengue case records, recent dengue case reports, location data using the Google Maps API, and weather data, including rainfall, temperature, and humidity, from PAGASA or Open-Meteo.

The next step in the process is the preprocessing of the collected data. In this stage, cleaning, integration, and preparation activities are carried out to ensure the data's accuracy, completeness, and consistency. Preprocessing includes detecting and correcting any missing values, duplicate records, or inconsistent data. The records of historical and current dengue cases, location data, and weather data are integrated to form one dataset compatible with machine learning algorithms. After completing this stage, the data will be consistent, reliable, and properly formatted for prediction.

The preprocessed data is then inputted into the Machine Learning Model. The system uses a Random Forest model implemented in the Scikit-learn Python library. Before making predictions, this model is trained using previous dengue case records and associated environmental variables, such as rainfall, temperature, humidity, and location. The Random Forest learns the relationships between these variables and previous cases of dengue during this training. After training, the model accepts preprocessed data and predicts possible dengue hotspot areas, classifying each location according to its level of risk.

The prepared data is sent to the Machine Learning Model, which uses the Random Forest Algorithm through the Scikit-learn library. First, the data is prepared and used to train the model using historical dengue case records. After the training is finished, the model studies the data and predicts which areas are likely to become dengue hotspots and identifies their risk level .

The predicted hotspot locations, with their respective risk levels, are input into the GIS-based hotspot mapping and visualization, integrating the predictions with their respective geographic coordinates. This module outputs the predicted hotspot areas in an interactive digital map, helping health officials quickly identify low, moderate, high-risk and critical communities in terms of dengue. Thus, this GIS visualization provides an intuitive spatial distribution of dengue, which makes monitoring and decision-making more effective. The GIS visualization provides a clear spatial representation of dengue distribution, making monitoring and decision-making more effective.

The mapped prediction results are finally used as input for the Prediction and Monitoring Output module. The system output, including the predicted dengue hotspot areas and their risk levels, is made available through DengueWatch Web and Mobile applications in the form of alerts, notifications, reports, and analytics. The CHO, CHWs, BHWs, LGUs, and other authorized users can track dengue trends in Iligan City, get early warning alerts, analyze reports, and make timely and appropriate decisions to strengthen their dengue prevention and control programs.

Finally, all prediction results, GIS hotspot maps, alerts, notifications, reports, and analytics are displayed through the DengueWatch web and mobile applications. These applications enable authorized users to monitor dengue trends, receive early warning alerts, analyze reports, and make timely decisions to strengthen dengue prevention and control programs in Iligan City.

System Design

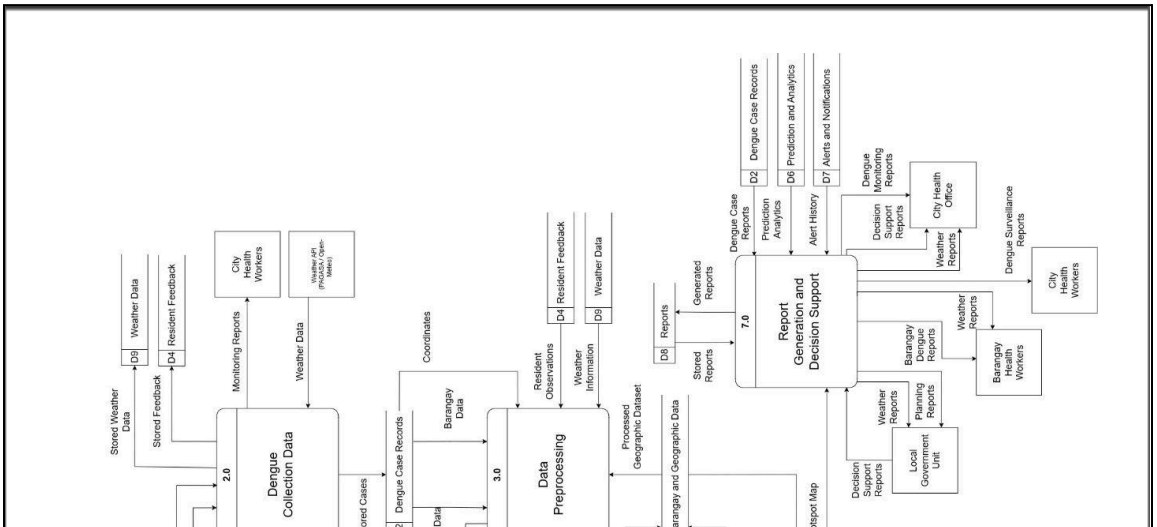


Figure 3.4 Data Flow Diagram

Figure 3.4 shows the Data Flow Diagram (DFD) of DengueWatch: An AI-Assisted Predictive Dengue Surveillance and Hotspot Analytics with Geographic Information System. The process begins with User Authentication (1.0), where residents, Barangay Health Workers (BHWs), City Health Workers (CHWs), and the City Health Office (CHO) register and log in to the system.

Their login credentials are verified using the User Accounts (D1) database. Once authenticated, users are granted access to the system according to their assigned roles.

After successful authentication, the process proceeds to Dengue Collection Data (2.0). In this stage, Barangay Health Workers submit dengue case reports, while residents provide observations or feedback related to possible mosquito breeding sites or dengue concerns. At the same time, the system collects weather data, including rainfall, temperature, and humidity, from the Weather API (PAGASA/Open-Meteo). The collected dengue case reports, resident feedback, and weather information are stored in the Dengue Case Records (D2), Resident Feedback (D4), and Weather Data (D9) databases. Monitoring reports are also made available to City Health Workers.

Next, the collected data is sent to Data Preprocessing (3.0). During this stage, the system retrieves dengue case records, resident feedback, barangay and geographic data, and weather information. The data is cleaned, checked for missing or duplicate records, integrated, and prepared for analysis. The processed information is then stored in the Integrated Dengue and Weather Dataset (D5), which serves as the main dataset for machine learning.

The integrated dataset is then forwarded to AI-Based Dengue Hotspot Prediction (4.0). At this stage, the system applies the Random Forest algorithm using the Scikit-learn library to analyze historical dengue records, current

dengue case reports, geographic information, demographic data, and weather variables such as rainfall, temperature, and humidity. Based on the identified patterns, the model predicts possible dengue hotspot areas and determines the corresponding risk levels. The prediction results are stored in the Prediction and Analytics (D6) database.

After the prediction process, the results are sent to Hotspot Mapping and Visualization (5.0). This module combines the prediction results with the barangay geographic data through the Geographic Information System (GIS) to generate interactive hotspot maps showing the locations and severity of dengue risk. These maps are accessible to the City Health Office, Barangay Health Workers, Local Government Units (LGUs), and residents for monitoring and decision-making.

The system then proceeds to Alert and Notification Management (6.0). Based on the predicted hotspot areas and risk levels, the system generates dengue alerts and warning notifications. These alerts are stored in the Alerts and Notifications (D7) database and are automatically sent to residents, Barangay Health Workers, Local Government Units, and City Health Workers to support timely response and dengue prevention activities.

Finally, the process ends with Report Generation and Decision Support (7.0). The system gathers information from the Dengue Case Records (D2), Prediction and Analytics (D6), Alerts and Notifications (D7), Weather Data (D9),

and geographic datasets to generate reports such as dengue case reports, hotspot analysis reports, weather reports, surveillance reports, planning reports, and decision support reports. These reports help health authorities and local government officials monitor dengue trends, evaluate hotspot predictions, assess the influence of weather conditions, and make informed decisions for effective dengue prevention and control in Iligan City.

Process Flow Diagram

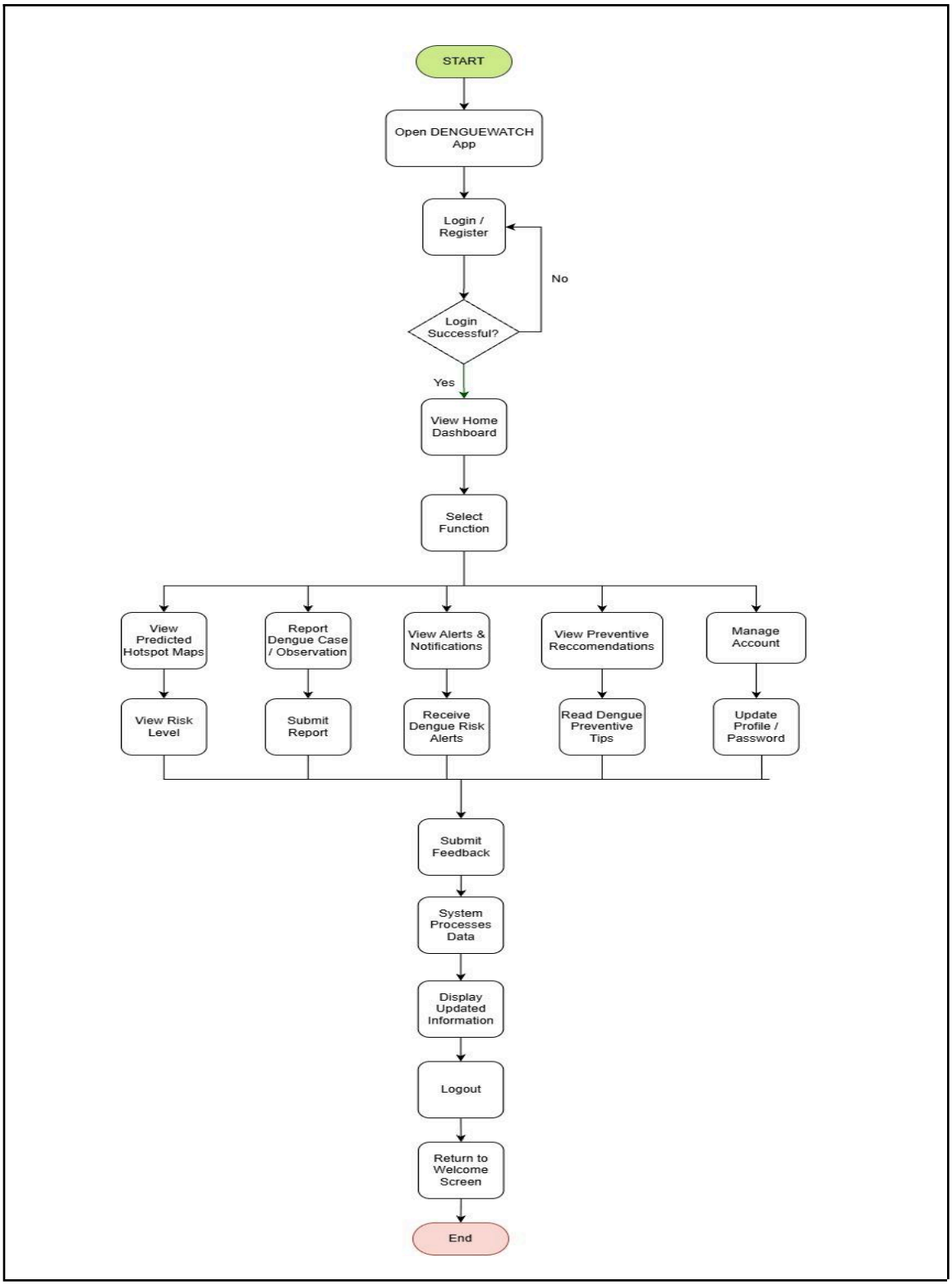


Figure 3.5 Resident Process Flow Diagram

Figure 3.5 shows how residents use the DengueWatch mobile application. The process begins when a resident opens the app and logs in or creates a new account. If the login details are incorrect, the resident is asked to try again until the correct credentials are entered. The resident is directed to the Home Dashboard, where they can use the application's primary functions, following a successful login.

Residents can assess the amount of risk in their neighborhood and explore maps of dengue hotspots from the dashboard. Additionally, they can read the system's prevention advice and suggestions, report dengue-related observations, and get alerts and updates about potential dengue hazards. These characteristics assist locals in remaining knowledgeable and conscious of the dengue situation in their neighborhood. Additionally, the application enables users to modify their passwords and profile details as needed.

Residents can leave comments on the program once they have used all of its features. After processing the submitted data, the system updates the relevant information. The process concludes when the resident logs out of the program and returns to the welcome screen. Residents can direct participate in dengue monitoring, stay informed about the dengue risks, and support community-wide dengue prevention programs through this procedure.

System Design

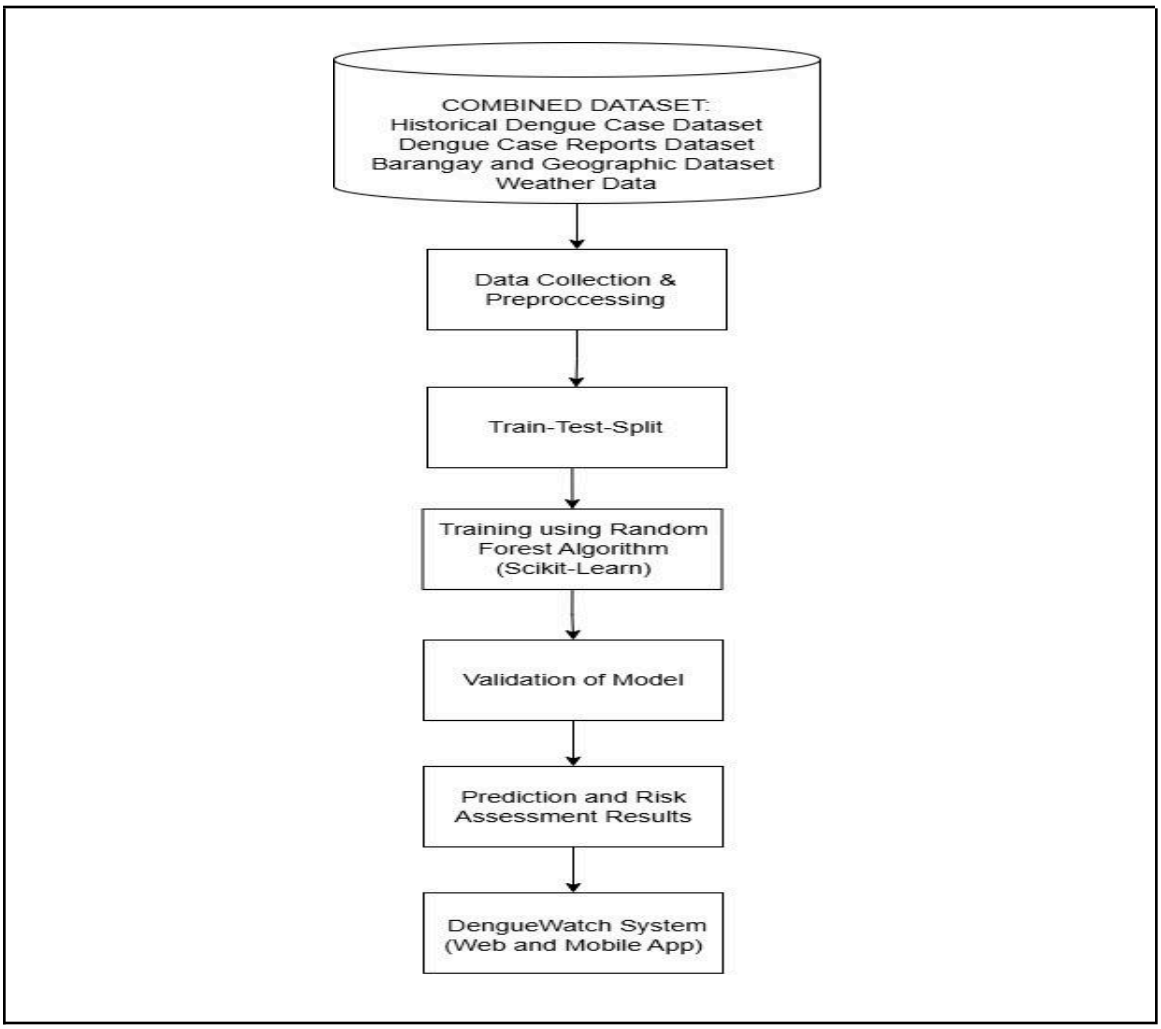


Figure 3.6 Process Flow Chart

The DengueWatch: An AI-Assisted Predictive Surveillance and Hotspot Analytics with Geographic Information System design is shown in Figure 3.4. The process by which the system gathers data, processes it, trains a prediction model, and provides the findings to end users via the web and mobile application is depicted in detail in the image.

1. Combined Dataset

The Combined Dataset, which combines data from different sources used in the study, is where the process starts. Among these datasets are:

- Historical Dengue Case Dataset – includes information about past dengue cases in Iligan City.
- Dengue Case Reports Dataset – includes recently submitted dengue case reports from healthcare workers.
- Barangay and Geographic Dataset – includes information about the locations of Iligan City's barangays and puroks.
- Weather Data - contains environmental information such as rainfall, temperature, and humidity obtained from PAGASA or Open-Meteo.

These datasets are combined to create a single dataset that is used to train and predict models.

2. Data Collection and Preprocessing

The system collects and preprocesses the data after merging the datasets. Incomplete, redundant, and unnecessary records are eliminated by cleaning, validating, and organizing the gathered data. Dengue cases are subsequently categorized into various risk levels using data labeling or risk classification.

The risk levels are classified as follows:

- Low Risk: 1–5 reported cases
- Moderate Risk: 6–50 reported cases
- High Risk: 51 - 100 reported cases
- Critical: More than 101 reported cases

According to the City Health Office's suggestions and past dengue records may be used to modify these levels. Data correctness, consistency, and machine learning applicability are guaranteed by this procedure.

3. Train-Test Split

The preprocessed dataset is split using the train-test split method into training and testing sets prior to model training. The machine learning model's performance is assessed using the testing dataset, while it is trained using the training dataset. This method aids in assessing the model's capacity for generalization and prediction on unobserved data.

4. Training Using the Random Forest Algorithm (Scikit-Learn)

The system moves on to model training once the dataset is ready. To find possible dengue hotspot sites, the Random Forest Algorithm—which is implemented using the Scikit-Learn library—is trained using historical dengue cases, geographic location data, and weather data such as rainfall, temperature, and humidity.

The algorithm may anticipate possible dengue hotspot areas and associated risk levels by using this machine learning technique to find correlations between dengue cases and geographic locations. Because Random Forest can handle complicated datasets and deliver dependable prediction performance, it was selected.

5. Validation of Model

The model's prediction accuracy and dependability are assessed through validation after training. The model's performance is assessed using suitable evaluation measures on the testing dataset.

The preprocessing and training phases are repeated until appropriate outcomes are obtained if the model's performance is inadequate. The system uses the model for prediction only when it satisfies the necessary degree of accuracy and dependability.

6. Prediction and Risk Assessment Results

The system produces prediction and risk assessment findings based on successful validation, such as:

- Iligan City's predicted dengue hotspot locations.
- There are four levels of dengue risk: low, moderate, high and critical.
- Messages and alerts about dengue for those who are worried.

- Analytics and statistical information to assist prompt decision-making and dengue prevention initiatives.

These outputs provide a health authorities and residents with the information that can help reduce the spread of dengue and improve response strategies.

7. DengueWatch System (Web and Mobile Application)

Lastly, the DengueWatch mobile and web application incorporates the prediction results. Depending on their positions, different users can access different system functions:

- City Health Office (CHO) – manages the system, monitors dengue cases, views hotspot predictions, generates reports, and oversees dengue surveillance.
- City Health Workers (CHWs) – submit and manage dengue case reports, monitor hotspot predictions, and access reports.
- Barangay Health Workers (BHWs) – report dengue cases within their assigned barangays, monitor local hotspot areas, and assist in dengue surveillance.
- Local Government Units (LGUs) – access hotspot maps, reports, and analytics to support planning and dengue prevention activities.

- Residents – use the mobile application to view dengue hotspot maps, receive risk alerts and preventive reminders, and stay informed about dengue conditions in their communities.

Training Procedure

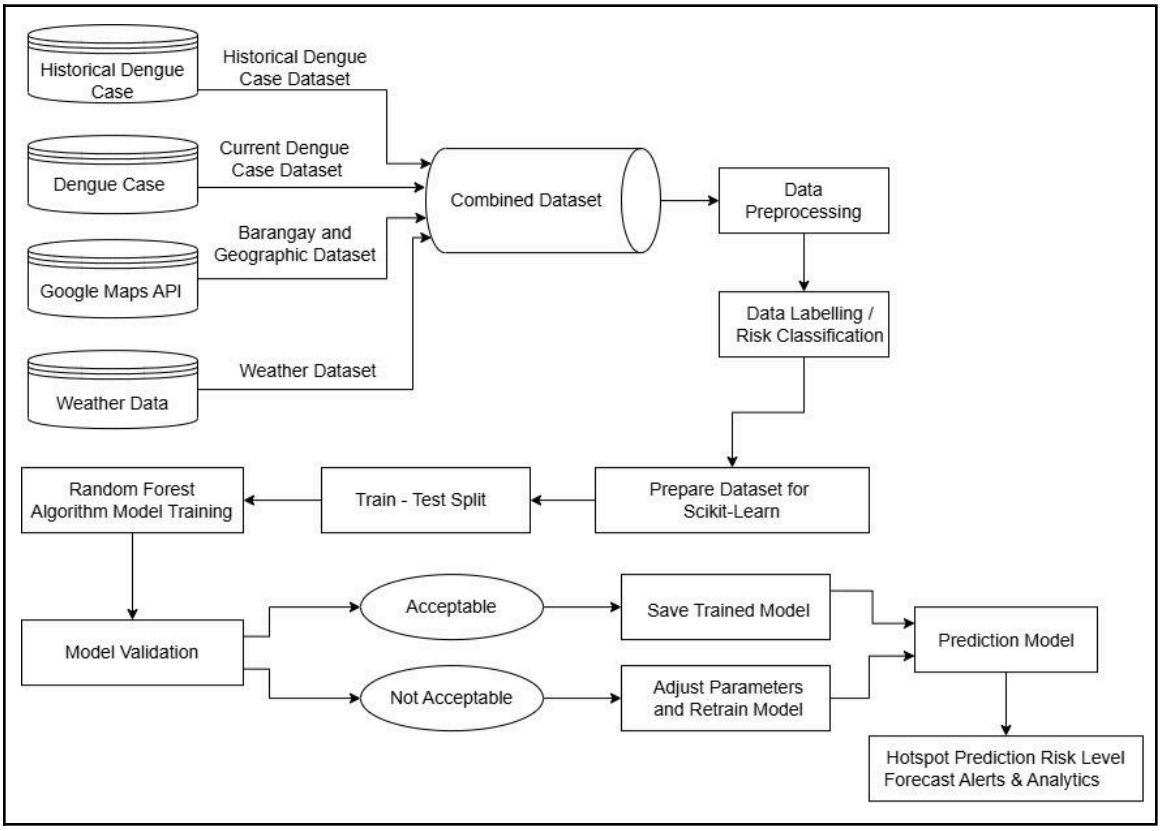


Figure 3.7 Training Procedure

Figure 3.7 illustrates the model training procedure used to develop the DengueWatch system. The process begins with four datasets. The Historical Dengue Case Dataset contains records of previous dengue cases in Iligan City. The Current Dengue Case Dataset consists of newly reported dengue cases submitted by health workers. The Barangay and Geographic Dataset contains the geographic locations and coordinates of barangays and puroks obtained through

the Google Maps API. The Weather Dataset includes environmental information such as rainfall, temperature, and humidity collected from PAGASA or Open-Meteo. These datasets are combined into a single dataset to provide complete information for model training.

After the datasets are merged, the system performs Data Preprocessing, where the collected data is cleaned, validated, and organized by removing incomplete, duplicate, and inconsistent records. This step ensures that the dataset is accurate, consistent, and suitable for machine learning.

Following preprocessing, the system performs Data Labelling or Risk Classification, where dengue cases are categorized into three risk levels based on the number of reported cases. The risk levels are defined as follows:

Risk Level	Number of Cases
Low Risk	1-5
Moderate Risk	6-50
High Risk	51-100
Critical	101 above

According to the City Health Office (CHO), the following risk classifications are used as labels for the machine learning model. Low Risk (1–5 cases) indicates a low number of dengue cases in an area. Moderate Risk (6–50 cases) requires closer monitoring and early prevention measures. High Risk

(51–100 cases) indicates a higher risk of an outbreak and requires immediate attention. Critical Risk (more than 101 cases) indicates a serious situation that needs urgent intervention. These classifications help DengueWatch identify areas based on their risk levels, allowing health authorities to prioritize monitoring and take appropriate actions to prevent the spread of dengue.

Feature	Possible Values
Age	< 1 ,1-4 , 5-14, 15-24, 25-49, 50-59, 60+
Sex	Male, Female
Blood Type	A+, A-, B+, B-, AB+, AB-, O+, O-

According to the National Clinical Guideline of Dengue for Timor-Leste, which follows World Health Organization standards, dengue cases are grouped using the age brackets < 1, 1-4, 5-14, 15-24, 25-49, 50-59, and 60+ instead of general population age groups, because dengue affects age groups differently. Infants under 1 year old still carry some protection from their mother's antibodies, while the 1-4 and 5-14 groups are the most affected, making up 33.5% and 37.8% of cases in Timor-Leste. The 15-24 group shows dengue spreading within communities, the 25-49 group covers working adults exposed through daily activities, and the 50-59 and 60+ groups are separated since older adults face higher risk of severe illness or death. Using this age grouping helps the DengueWatch system and its Random Forest model make more accurate predictions of dengue hotspots in Iligan City [36].

The Age, Sex, and Blood Type variables are encoded together with historical dengue records, current dengue reports, geographic information, and weather data such as rainfall, temperature, and humidity. These features enable the Random Forest algorithm to identify patterns associated with dengue occurrence and improve the prediction of possible hotspot areas in Iligan City.

After data labeling, the dataset is prepared for the Scikit-Learn library by converting it into a format compatible with the Random Forest algorithm. The prepared dataset is then divided using the Train-Test Split method. One portion of the data is used for training the model, while the remaining portion is used for testing and validation to evaluate the model's performance on unseen data.

The training dataset is then used for Random Forest Algorithm Model Training. During this stage, the Random Forest model learns the relationships among historical dengue cases, demographic information, geographic locations, and weather conditions. These learned patterns enable the model to predict possible dengue hotspot areas and classify their corresponding risk levels.

After training, the system performs Model Validation to evaluate the accuracy and reliability of the trained model. If the model achieves acceptable performance, it proceeds to the Save Trained Model stage, where the trained model is stored and deployed as the Prediction Model. However, if the model does not achieve satisfactory performance, the system adjusts the Random Forest

parameters and retrains the model. This process is repeated until the model reaches an acceptable level of prediction accuracy.

Finally, the Prediction Model generates the system outputs, including predicted dengue hotspot areas, risk level classifications, alerts and notifications, and reports and analytics. These outputs are displayed in the DengueWatch web and mobile application, allowing the City Health Office (CHO), City Health Workers (CHWs), Barangay Health Workers (BHWs), Local Government Units (LGUs), and residents to monitor dengue cases, identify high-risk areas, and support timely dengue prevention and control efforts in Iligan City.

Entity Relationship Diagram

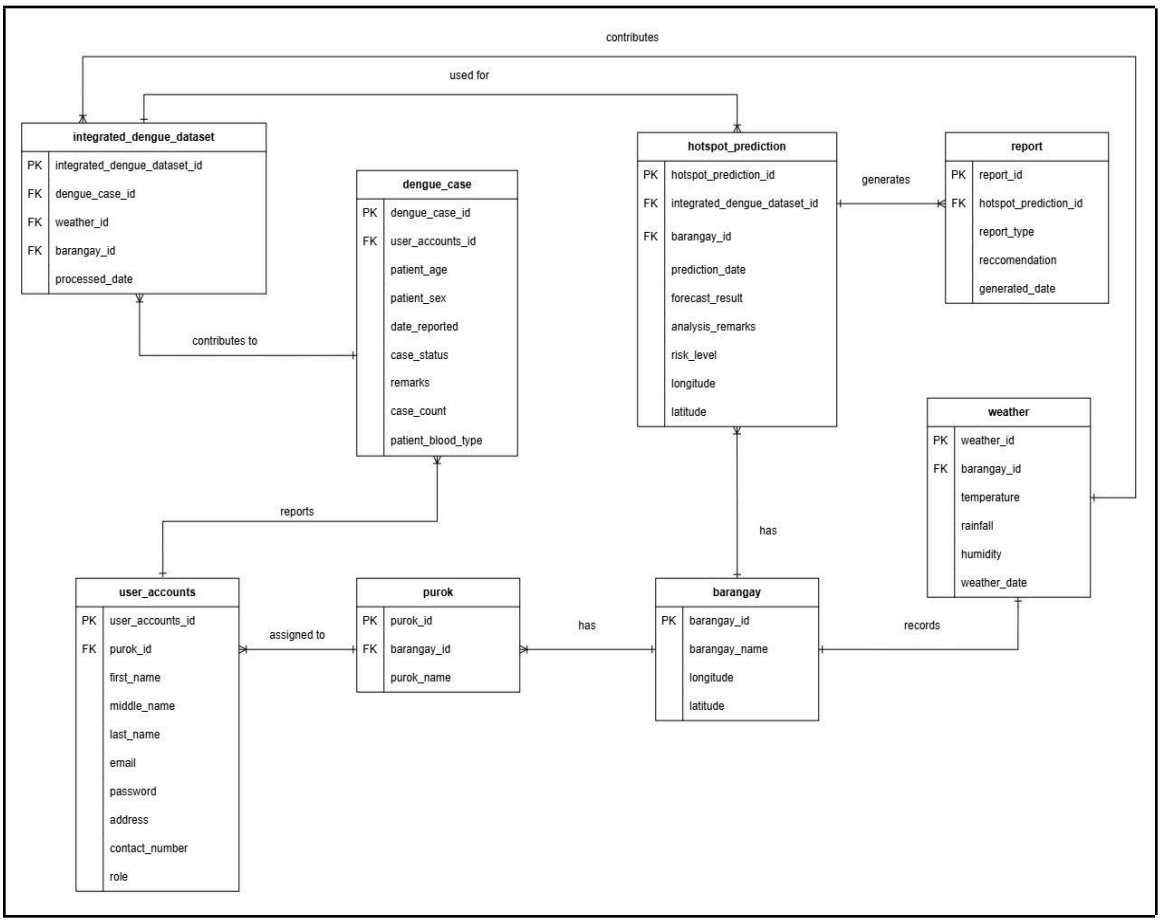


Figure 3.8 Entity Relationship Diagram

Figure 3.8 shows the database structure and the relationships between the entities used by the proposed DengueWatch: An AI-Assisted Predictive Dengue Surveillance and Hotspot Analytics with Geographic Information System (GIS). It presents the database tables used in the system and shows how these tables are connected to store and manage dengue-related information.

The User Accounts table stores the information of all registered users, including residents, Barangay Health Workers (BHWs), City Health Workers

(CHWs), and City Health Office (CHO) personnel. Each user is assigned to a specific Purok, and health workers can record dengue cases in the system.

The Purok table stores the list of puroks in Iligan City. Each purok belongs to one Barangay, allowing the system to identify the location of dengue cases.

The Barangay table stores the names and geographic coordinates (latitude and longitude) of barangays. These coordinates are used by the GIS feature to display dengue hotspot locations on the map. The barangay table is also connected to the Weather and Hotspot Prediction tables.

The Dengue Case table stores information about reported dengue cases, such as the patient's age, sex, blood type, date reported, case status, remarks, and case count. Each dengue case is linked to the user who recorded it.

The Weather table stores weather information, including temperature, rainfall, humidity, and the weather date. Each weather record is linked to a barangay and is used together with dengue case records to improve hotspot prediction.

The Integrated Dengue Dataset table stores the processed data after combining dengue case records, weather data, and barangay information. This processed dataset is used as the input for the Random Forest machine learning model.

The Hotspot Prediction table stores the prediction results generated by the Random Forest model. It includes the prediction date, prediction result, risk level, and analysis remarks. Each prediction is linked to a barangay and the processed dataset used during prediction.

Finally, the Report table stores the reports generated by the system. It contains the report type, recommendations, and the date the report was created. These reports help the City Health Office, Barangay Health Workers, and Local Government Units monitor dengue cases, identify hotspot areas, and support decision-making for dengue prevention and control.

Data Dictionary

In this phase, the researchers had constructed a data dictionary in order to have a collection of descriptions of the data objects or items in a data model for the benefit of programmers and others who need to refer to them. It consists of different tables, fields, descriptions, and types of data used in the development of the system.

Table 3.1
User Accounts

Attributes	Type	Length	Description
user_accounts_id	int	11	Unique identifier for each user.
purok_id	int	11	References the purok where the user belongs.
first_name	varchar	50	User's first name.
middle_name	varchar	50	User's middle name.
last_name	varcher	50	User's last name.
email	varchar	100	User;s email address used for login.
password	varchar	255	Encrypted password of the user.
address	varchar	255	Complete residential address.
contact_number	varchar	15	User's contact number.
role	varchar	30	User role (Resident, Barangay Health Workers, City Health Workers, City Health Office, and Local Government Units).

Table 3.1 User Accounts table stores the personal information and login credentials of all users registered in the DENGUEWATCH system. This includes residents, Barangay Health Workers, City Health Workers, and City Health Office personnel. It contains the user's basic information, contact details, password, and assigned role. The table is used for user authentication and authorization, ensuring that only authorized users can access the system according to their

assigned responsibilities. It is also linked to the Purok table to identify the user's location.

Table 3.2
Barangay

Attributes	Type	Length	Description
barangay_id	int	11	Primary key. Unique identifier for each barangay.
barangay_name	varchar	50	Official name of the barangay.
longitude	decimal	10,7	Geographic longitude coordinate of the barangay center.
latitude	decimal	10,7	Geographic latitude coordinate of the barangay center.

Table 3.2 the Barangay table stores the geographic information of all barangays included in the system. It contains the barangay name together with its latitude and longitude coordinates, which are used by the Geographic Information System (GIS) for mapping and visualization. This table allows the system to display dengue hotspot locations accurately on an interactive map and helps health authorities identify barangays with higher dengue risk.

Table 3.3
Purok

Attributes	Type	Length	Description
purok_id	int	11	Primary key. Unique identifier for each purok.
barangay_name	varchar	50	Official name of the barangay.
purok_name	varchar	100	Purok classification of the barangay within the city.

Table 3.3 the Purok table contains the list of puroks within each barangay. Each purok is linked to a specific barangay through the barangay_id foreign key. This table helps the system organize users and dengue cases according to their exact location. By identifying the purok of each user or reported case, the system can provide more accurate spatial analysis and support hotspot identification at the community level.

Table 3.4
Dengue Case

Attributes	Type	Length	Description
dengue_case_id	int	11	Primary key. Unique identifier for each dengue case report.
user_accounts_id	int	10	Foreign key referencing the user who reported the case.
barangay_id	varchar	11	Foreign key referencing barangay. Identifies the barangay where the case was reported.
patient_age	int	3	Age of the dengue patient at the time of reporting.
patient_sex	varchar	255	Biological sex of the patient: male or female.
patient_blood_type	varchar	5	Blood type of the patient.
date_reported	date	11	Date of the dengue case that was officially reported into the system.
case_status	enum	-	Current status of the case: suspected, confirmed, recovered, or deceased.
remarks	text	100	Additional notes or observations from the dengue case.
case_count	int	10	Number of dengue cases reported.

Table 3.4 the Dengue Case table stores all reported dengue case information used by the system. It includes the reporting user, patient information such as age and sex, the date the case was reported, case status, remarks, and the total number of reported cases. This table serves as one of the primary sources of data for the AI-assisted prediction model. The information stored in this table is used during data preprocessing and machine learning to identify patterns and predict possible dengue hotspot areas.

Table 3.5
Weather

Attributes	Type	Length	Description
weather_id	int	11	Primary key. Unique identifier for each weather record.
barangay_id	int	11	Foreign key referencing the barangay connected to the weather record.
temperature	decimal	5,2	Recorded temperature in degrees Celsius.
rainfall	decimal	8,2	Recorded rainfall amount in millimeters.
humidity	decimal	5,2	Recorded relative humidity percentage.
weather_date	date	-	Date when the weather information was recorded.

Table 3.5 the Weather table stores rainfall, temperature, and humidity data collected from PAGASA or Open-Meteo. Each weather record is connected to a barangay and contributes to the integrated dataset used for dengue prediction.

Table 3.6
Integrated Dengue Dataset

Attributes	Type	Length	Description
Integrated_dengue_ dataset_id	int	11	Primary key. Unique identifier for each integrated dataset record.
dengue_case_id	int	11	Foreign key referencing dengue case data.
weather_id	int	11	Foreign key referencing the weather record included in the dataset.
barangay_id	int	11	Foreign key referencing the barangay connected to the integrated record.
processed _date	datetime	-	Date and time when the dataset was processed and integrated.

Table 3.6 the Integrated Dengue Dataset table stores the processed information after dengue case records, weather data, and barangay information are cleaned and combined. This table serves as the main input for the Random Forest prediction model.

Table 3.7
Hotspot Prediction

Attributes	Type	Length	Description
hotspot_prediction_id	int	11	Primary key. Unique identifier for each prediction record.
integrated_dengue_dataset_id	int	11	Foreign key referencing the integrated dataset used for prediction.
barangay_id	int	11	Foreign key referencing barangay. The barangay for which the prediction was generated.
prediction_date	date	-	The date for which the dengue risk is being predicted.
forecast_result	text	255	Detailed narrative or summary of the prediction outcome and supporting data.
analysis_remarks	text	255	Additional remarks or analysis regarding the prediction.
analysis_remarks	text	255	Additional remarks or analysis regarding the prediction.
longitude	decimal	10.7	Longitude of the predicted hotspot.
latitude	decimal	10.7	Latitude of the predicted hotspot.
risk_level	enum	-	Risk Classification (Low, Moderate, High and Critical).

Table 3.7 the Hotspot Prediction table stores the prediction results produced by the Random Forest model. Each prediction is connected to an integrated dataset and a specific barangay. The longitude and latitude values are used to display hotspot on the GIS map.

Table 3.8
Report

Attributes	Type	Length	Description
report_id	int	11	Primary key. Unique identifier for each generated report.
hotspot_prediction_id	int	11	Foreign key referencing hotspot_prediction. Links the report to the prediction it documents.
report_type	enum	-	Classification of the report: weekly, monthly, quarterly, or annual.
generated_date	datetime	-	Date and time the report was generated by the system.
recommendation	text	500	Health interventions or action recommendations included in the report.

Table 3.8 the Report table stores the reports generated from the hotspot prediction results. Each report is linked to a specific hotspot prediction and contains information such as the report type, recommendations, and the date the report was generated. The reports provide summarized information and recommended actions that support decision-making for dengue monitoring, prevention, and response. These reports can be accessed by the City Health Office, Barangay Health Workers, and other authorized users to assist in planning and implementing dengue control measures.

System Requirements

Hardware Specifications

The researchers will use these hardware specifications for developing the DengueWatch: An AI-Assisted Predictive Dengue Surveillance and Hotspot Analytics with Geographic Information System.

Development Hardware

For development, the researchers will use a Ryzen 5 5600X processor with 16 GB of RAM and 1TB of storage. A 23.8-inch IPS monitor with a 100Hz refresh rate will be used, along with a keyboard, mouse, and a stable internet connection. The operating system used for development is Windows 11.

Deployment Hardware

For deployment, a laptop or desktop computer should be enough to use the web version of the system, while a smartphone running Android 8.0 or a later version can be used to access the mobile application. A stable internet connection is also needed for both the web and mobile applications to work properly, especially for real-time monitoring, reporting, and receiving alerts.

Software Specifications

The researchers used these software tools and technologies to develop the system.

Development Software

The system will be built using Python 3.12 and a framework called Django as the main programming language and framework, together with Scikit-learn, a Python library used for implementing the Random Forest algorithm for dengue hotspot prediction. The researchers also used Visual Studio Code as the code editor, GitHub for version control, Firebase (Cloud Firestore) as the cloud-based database, the Google Maps API for location and geographic data, and an SMS API for sending alerts and notifications to users.

Deployment Software

For deployment, the system requires a computer running Windows 10 or 11, or a smartphone running Android 8.0 or a later version. A web browser such as Google Chrome or Microsoft Edge is needed to access the web application, along with an SMS gateway for sending alert notifications and a stable internet connection for real-time monitoring and reporting.

Table 3.8
Respondents of the Study

Respondents	Frequency	Percentage
City Health Office	4	8%
City Health Workers	8	16%
Barangay Health Workers	8	16%
Local Government Unit	6	12%
Residents	24	48%
Total	50	100%

Performance Metrics

The study will employ usability testing to evaluate the DengueWatch system in terms of effectiveness, efficiency, and user satisfaction. This type of testing is appropriate because the system will be used by multiple stakeholders, including the City Health Office (CHO), City Health Workers (CHWs), Barangay Health Workers (BHWs), Local Government Unit (LGU) representatives, and residents. The system therefore has to be functional, reliable and easy to use.

The technique shall be conducted in the following stages:

Preparation

1. The researchers will prepare DengueWatch system in its final version with web and mobile applications.
2. Test accounts will be generated per user role like City Health Office, City Health Workers, Barangay Health Workers, Local Government Unit Representatives and Residents.

3. A usability testing plan will be developed to lead the participants during testing containing task scenarios and evaluation questionnaires.
4. Prior to the testing, the researchers will confirm that all the functions of the system are functioning properly – dengue case reporting, hotspot mapping, dengue risk prediction, alert messages, report generation – so that the evaluation process goes on smoothly.

Orientation

1. The participants will be given a short briefing about the study's objectives and the purpose of the usability testing.
2. The researchers will describe the tasks that the participants have to do with the system.
3. Participants will be advised that the testing is for the purpose of evaluating the system, not their own talents or performance.
4. Directions will be supplied to you on how to access and navigate the DengueWatch system.

Task Execution

1. Participants will check in with their given user account.

2. Barangay Health Workers and City Health Workers will make information on dengue cases and monitor data on dengue.
3. City Health Office personnel and LGU representatives will pilot the functionalities for hotspot monitoring, analytics dashboards and report generating.
4. The residents will be able to test the mobile application having maps of hotspots, status of dengue risk, notifications and feedback/reports.
5. The researchers will record any problems, mistakes or issues that participants experience when completing the tasks assigned to them.

Observation and Recording

1. Researchers will study how users interact with the DengueWatch system when they are executing their assigned responsibilities of submitting dengue case reports, viewing hotspot maps, checking dengue risk levels, creating reports, and receiving alert alerts.
2. Participants' errors, delays, navigation problems and system-related problems will be noted and documented during the exam.
3. Notes, screenshots and system logs can be collected to help identify usability difficulties and areas for development.
4. The researchers will record participant feedback and observations on the functionality, usability, and overall experience of using the DengueWatch web and mobile applications.

5. All collected observations and recordings will be utilized to evaluate the system's efficacy, efficiency and user happiness, and for further system improvement.

Feedback Collection

1. After completing the tasks, users will answer the **System Usability Scale (SUS)** questionnaire to evaluate the usability of the DengueWatch system.
2. User happiness, usability and general experience with the system can be assessed by a brief interview or feedback form to collect further comments and ideas.

Data Analysis

1. The data obtained through observations, feedback forms and SUS questionnaire will be organized, tallied and analyzed.
2. The results will be interpreted in terms of the following usability attributes:

   a. **Effectiveness** - the percentage of participants who effectively complete their assigned tasks utilizing the DengueWatch system.

   b. **Efficiency** – based on the average time taken and the amount of mistakes found when doing tasks including submitting dengue case reports, monitoring hotspot regions, generating reports and obtaining notifications.

c. **Satisfaction –** determined by System Usability Scale (SUS) scores and feedback from the participants evaluating their overall experience and acceptance of the system.

3. The findings will provide insights into the usefulness and acceptance of the DengueWatch system and provide directions for future development.

Evaluation Tool

Researchers will evaluate the DengueWatch system by using a survey questionnaire based on the System Usability Scale (SUS) and a 5-point Likert

PScale. The scale ratings shall be read as follows; 5 – Excellent, 4 – Good, 3 – Fair, 2 – Poor and 1 – Very Poor.

The evaluation participants will consist of representatives comprising a total of 50 respondents, which include 5 City Health Office Staff, 5 City Health Workers, 10 Barangay Health Workers, 5 Local Government Unit representatives, and 25 Iligan City residents. The DengueWatch online and mobile applications will be assessed by the participants after usage based on functionality, usability, effectiveness, efficiency and overall user satisfaction.

The collected feedback will be used to evaluate the system's acceptability and usability, as well as to determine areas for future improvement.

Table 3.9
System Usability Scale (SUS) Score Interpretation Table

Rating	Scoring Scale	Qualitative Interpretation
5	4.20 - 5.00	Very Acceptable
4	3.40 - 4.19	Acceptable
3	2.60 - 3.39	Fairly Acceptable
2	1.80 - 2.59	Slightly Acceptable
1	1.00 - 1.79	Not Acceptable

Table 3.9 System Usability Scale (SUS) Score Interpretation Table is intended to assess the overall usability and acceptability of the system. The average ratings calculated are categorized into five degrees of the usability. A

mean score of 4.20 to 5.00 is Very Acceptable, indicating great usability and a very high level of user satisfaction. Scores between 3.40 to 4.19 are acceptable and indicate that the system is working properly, with only minor usability issues. The range of mean scores from 2.60 to 3.39 is Fairly Acceptable and represents the average level of usability and user satisfaction. Scores from 1.80 to 2.59 are read as Slightly Acceptable, which means that the system has several usability difficulties that may need to be improved. Finally, mean ratings ranging from 1.00 to 1.79 are categorized as Not Acceptable, signifying inadequate usability, low user satisfaction, and a requirement for major system upgrades.

Ethical Considerations

The DengueWatch: An AI-Assisted Predictive Dengue Surveillance and Hotspot Analytics with Geographic Information System is executed to ensure privacy, safety and rights of all participants. The study will comply with the Data Privacy Act of 2012 to protect personal information and collected data .

The system will restrict access to authorized users via secured login accounts and role-based access. Participants will be told about the purpose of the study before evaluation. Participation will be voluntary and they can leave at any time with no penalty. The personal information and other data regarding dengue obtained throughout the study will be kept confidential and used only for research reasons. The data will be maintained securely and only accessed by the researchers and authorised advisers. The system would incorporate security features including user authentication, access control and data encryption to secure user information. Participants will not be identified in any reports or publishing of information.

The researchers will conduct the study with fairness, honesty, transparency and respect for all volunteers. Ethical approval will be obtained before the commencement of the evaluation. Through these measures, the study aims to support better dengue monitoring and prediction while protecting the rights and privacy of all participants.

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ADVISER-BASED REVISIONS

DengueWatch: An AI-Assisted Predictive Dengue Surveillance and Hotspot Analytics with Geographic Information System

How to apply this insert. These red passages are replacement or supplemental material mapped to the printed page references in the adviser comments. Insert each passage into its corresponding chapter and remove conflicting statements from the original draft. The supplied manuscript is a PDF, so this copy preserves it and appends the red insert pages for placement in an editable manuscript.

1. Define the prediction target and horizon

Replace general statements such as “predict possible hotspots” with the following operational definition:

Prediction target. At the end of calendar month t, using only dengue records, weather observations, and other information available by the forecast issue cutoff, DengueWatch estimates the number of recorded dengue cases in calendar month t+1 for each of Iligan City’s 44 barangays. The primary model target is this next-month barangay case count. A barangay is flagged as a potential planning hotspot when its estimated count is at least 6. This is an internal planning threshold, not an official outbreak declaration or a prediction of an individual resident’s health status.

If the current month is still underway, the system may display a provisional estimate based on confirmed records received to date; the issue date and data cutoff must be shown. Actual weather or case information that became available after the issue cutoff must not be used as a predictor. The target is the following month’s observed count, not a risk label derived from the same month’s input count.

2. Use one documented set of risk bands

Use these bands consistently for the forecast map, analytics, and alert review. They describe internal planning categories for a predicted barangay-month count; they are not represented as CHO or DOH outbreak classifications unless a cited written guideline is obtained.

Predicted count per barangay-month	Display band	Interpretation
0-5	Low	Below the prototype planning threshold.
6-50	Moderate	Meets the internal 6-case planning flag.
51-100	High	Higher planning priority; not a clinical classification.
101 or more	Critical	Highest planning band; not an outbreak declaration.
Incomplete or unverified report	Unavailable	Do not convert missing reporting into zero or assign a risk band.

Apply the same boundary rules to forecast outputs only when describing predictions, and label historical case totals separately from predicted totals. If CHO provides a written official threshold, cite it and update every occurrence consistently.

3. Data feasibility and inventory

The current prototype inventory supports a monthly modeling panel. It does not yet establish that every historical month was completely reported or that the researchers have continuing authorization to use CHO records. The following figures describe the workbook-derived prototype inventory and must be reconfirmed with the data custodian before final claims are made.

Inventory item	Current prototype inventory
Source and record unit	Workbook-derived individual case line list; each source row represents a case record, not an aggregate count.
Coverage and geography	Admission dates from 2020-01-01 through 2026-07-31; official labels for 44 Iligan City barangays.
Source rows and usable rows	10,484 source rows; 10,296 have both a valid admission date and a matched official barangay.
Known quality gaps	133 rows have a missing/invalid admission date; 55 lack a matched official barangay; 163 potential duplicate matches are retained and flagged because no stable case identifier is available.
Reporting delay and case definition	Cannot be calculated from the supplied file: it has an admission date but no received/entered date. The workbook does not state a versioned case definition or consistently establish confirmed versus suspected status.
Monthly modeling panel	3,476 rows: 44 barangays across 79 calendar months, January 2020-July 2026. A row with no source records is not proof of zero incidence or complete reporting.
Weather input	Cached Open-Meteo values at one Iligan City center point (8.228 N, 124.2452 E), applied citywide: monthly precipitation sum (mm), monthly mean temperature (degrees C), and mean available hourly relative humidity (%). This is not barangay-level weather.
Weather gaps	The cached series is missing February-July 2026. Validate the source and coverage against CHO/PAGASA needs before operational use.

Feasibility gate. Before model claims or operational use, document written CHO authorization, data custodian, permitted fields, secure transfer/storage, refresh schedule, case-definition version, and reporting-completeness procedure. Add received dates and a barangay-month completeness indicator. Confirmed zero requires a complete reporting period explicitly reported as zero; absent records remain missing/unverified.

The current design uses one modeling row per barangay per month because that is the grain supported by the available dates and prototype forecast. A weekly candidate panel may be evaluated later, but its no-record weeks must remain unverified and excluded until weekly completeness is confirmed. The proposed 50 respondents evaluate usability; they are not the training sample and do not establish model-data sufficiency.

4. Chronological validation and leakage controls

Replace the random train-test split and repeated tuning against the test set with the following protocol. All barangays from the same calendar month remain together in one temporal partition.

- **Training period:** February 2020-November 2023. Fit learned preprocessing and the Random Forest pipeline using training data only.
- **Validation period:** December 2023-February 2025. Select features and tune parameters using this period and training data only.
- **Final test:** forecast inputs from March 2025-June 2026, evaluated against target months April 2025-July 2026. Lock this period before model tuning and score it once after all decisions are frozen.
- At each forecast origin, use only records and weather information that would have been available at that issue cutoff. Exclude later case reports, later observed weather, and same-period target values from predictors.
- Fit imputation, encoding, scaling, feature selection, and other learned preprocessing within the training pipeline. Do not fit these steps on validation or final-test data.
- Compare the count model with a persistence baseline that carries forward the most recent observed monthly count. Report MAE and RMSE for the count forecast. If thresholded hotspot classes are separately evaluated, report per-class precision, recall, F1, and a confusion matrix.
- Do not repeatedly tune against the final test period. After its one-time evaluation is recorded, a deployment model may be refit on all available history; preserve the locked evaluation results and model version.

Current prototype holdout results

Measure	Random Forest	Persistence baseline	Interpretation
MAE (cases)	1.704	1.5298	Lower is better; Random Forest is 11.4% worse.
RMSE (cases)	2.5026	2.6541	Lower is better; Random Forest is better on this metric.

The current model does not outperform the persistence baseline on both count metrics. Therefore, describe it as experimental planning support only; usability scores do not establish prediction accuracy, and no public-health action should be based on an unreviewed forecast.

For time-ordered evaluation, ordinary random cross-validation can train on future observations and evaluate on earlier ones. Use a chronological split such as the documented scikit-learn TimeSeriesSplit approach where appropriate: scikit-learn TimeSeriesSplit documentation.

5. Revised objectives and evaluation

Replace the overlapping objectives and any claim that a usability study proves outbreak prevention with these measurable objectives:

- Evaluate the resident community-report and health-worker dengue-report workflows using task completion, data-entry errors, and time to complete defined scenarios. These measures describe prototype workflow performance; they do not by themselves prove reduced reporting delays in routine CHO operations.
- Evaluate one-calendar-month-ahead barangay case-count forecasts against later observations using the chronological holdout and persistence baseline. Report MAE and RMSE; add classification measures only if the binary or multi-class risk output is separately evaluated.
- Verify mapping, case-review, and alert functions using defined functional scenarios, including CHO review before a public risk alert is issued. Do not describe a model forecast as a confirmed dengue case or outbreak.
- Evaluate usability through task performance and correctly scored SUS. Measure increased awareness only if a separate pre/post awareness instrument is added. Treat reduced outbreaks as a possible longer-term benefit, not an outcome established by this prototype evaluation.

Correct System Usability Scale procedure

Administer the standard ten-item SUS using five response options. For each odd-numbered item (1, 3, 5, 7, 9), subtract 1 from the response. For each even-numbered item (2, 4, 6, 8, 10), subtract the response from 5. Sum the ten contributions and multiply by 2.5 to obtain a score from 0 to 100. Do not calculate SUS as the average of 1-5 responses or apply custom acceptability bands to SUS. Analyze a locally written functionality or acceptability questionnaire separately, label it as custom, and report its own scale.

Attach the actual questionnaire and task scenarios in the appendices. Record task completion, errors, and elapsed time separately from SUS.

Consistent purposive usability sample (50 participants)

Participant group	n	Purpose
CHO staff	5	Review, monitoring, and approval workflows.
CHW	5	Suspected-case submission and follow-up.
BHW	10	Barangay-level case and field workflows.
LGU representatives	5	Planning maps, reports, and analytics.
Residents	25	Mobile alerts, hotspot map, and community reporting.
Total	50	Usability sample only; not a model-training or accuracy sample.

6. Align the data dictionary and technical architecture

Describe the prototype that is actually configured. The development system uses Django and its ORM with SQLite. Do not describe Cloud Firestore as the primary store for relational case records unless it is selected and configured for the deployed system. A production database choice should be stated only after it is confirmed.

Apply these corrections throughout the architecture, ERD, and data dictionary:

- **Barangay and Purok:** use integer primary keys and matching integer foreign keys. Purok stores barangay_id as a foreign key; do not define the relationship only by a barangay name.
- **DengueCase:** one row represents one individual case record. Store aggregated counts in a separate barangay-month table or derived dataset, not as an unrestricted case_count on each individual row.
- **Diagnosis and outcome:** keep suspected/confirmed diagnostic classification separate from outcomes such as recovered/deceased.
- **CommunityReport:** resident environmental observations have a separate entity and review workflow. They are not dengue cases and do not enter case-count targets unless CHO verifies and records a case through the proper workflow.
- **HotspotPrediction:** define each field once and store barangay_id, forecast issue timestamp, input cutoff, target month, predicted count, derived display band, and model version. Remove repeated risk_level and analysis_remarks entries.
- **User passwords:** store only Django’s securely hashed password representation; do not describe passwords as recoverably encrypted.

The prediction is reproducible only when its forecast issue date, exact input cutoff, target period, and model version are retained. Public maps should display aggregated barangay-level results. Identifiable case records remain restricted to authorized health personnel through role-based access. Document CHO-approved retention, secure deletion, access logging, and incident procedures before production use.

7. Privacy, feature justification, and CHO review

Keep resident environmental observations separate from verified dengue case records. Require CHO review before a forecast-based alert is published to residents. A forecast is planning support, not an outbreak declaration or clinical assessment.

Age, sex, and blood type originate as patient-level attributes and must not be presented as individual risk factors. The initial forecast should omit blood type unless its availability and lawful use are confirmed and a feature-ablation comparison demonstrates value on the same untouched final test period. If it is retained only for exploratory analysis, use de-identified, lagged barangay-month aggregates and never expose individual values in forecast outputs.

Document how patient-level source records are transformed into aggregate barangay-month features, who can access identifiers, what is removed or retained, and how long data are held. The public-facing map must not expose names, addresses, or other patient identifiers.

8. Statistical, reference, and sample-count audit

The following corrections are needed before submission. They are written conservatively because the supplied PDFs do not include the source documents needed to verify every citation:

- On printed page 3, delete the comparison that labels a change from 338,983 to 62,313 as a 73% increase. Those displayed values do not support that direction or percentage. Restore a comparison only after confirming the same geographic scope, date window, and source table, then show the calculation.
- On printed page 20, reconcile the Acharjee et al. (2022) attribution with reference [32], which lists Sarma et al. (2020). Until the source and description are matched, remove or rewrite the paragraph; do not invent bibliographic details.
- On printed page 22, verify that the neural-network review description matches the title and authors listed in reference [35]. Remove or rewrite it until confirmed.
- On printed page 12, define DengueWatch as the prototype developed in this study. Do not cite a general dengue-information page as the source of the project’s own name.
- Use one respondent allocation everywhere: 5 CHO, 5 CHW, 10 BHW, 5 LGU representatives, and 25 residents (50 total). Clearly identify it as a purposive usability sample.

A complete claim-by-claim audit of all remaining citations requires checking each cited publication or official source. Confirm the author, title, year, URL/DOI, and in-text citation number against the source itself before stating that the audit is complete. The supplied files do not establish CHO data authorization or an approved retention schedule; describe both as pending until supporting records are available.

9. Editorial and figure corrections

Before final submission, complete the following consistency pass across the manuscript:

- Replace the sideways or cropped data-flow diagram with the landscape diagram on the following insert page; enlarge text in the architecture and ERD.
- Use sequential figure and table numbers, then update every cross-reference in the text. Remove repeated methodology paragraphs and duplicated field definitions.
- Correct “AI-Asssited” to “AI-Assisted” and “breed more faster” to “breed more rapidly.”
- Use future tense for planned procedures and past tense only for work already completed. Recheck page breaks and table layouts after replacing diagrams and revising appendices.
- Use the following replacement workflow and conceptual ERD only after reconciling table and key names with the final Django models.

The pages that follow are replacement diagrams for the system workflow and core data relationships. Keep the diagram labels red as revised content; borders and connectors are visual guides.

Figure: revised system data flow

One barangay-month row; forecast the next calendar month using information available at issue time.

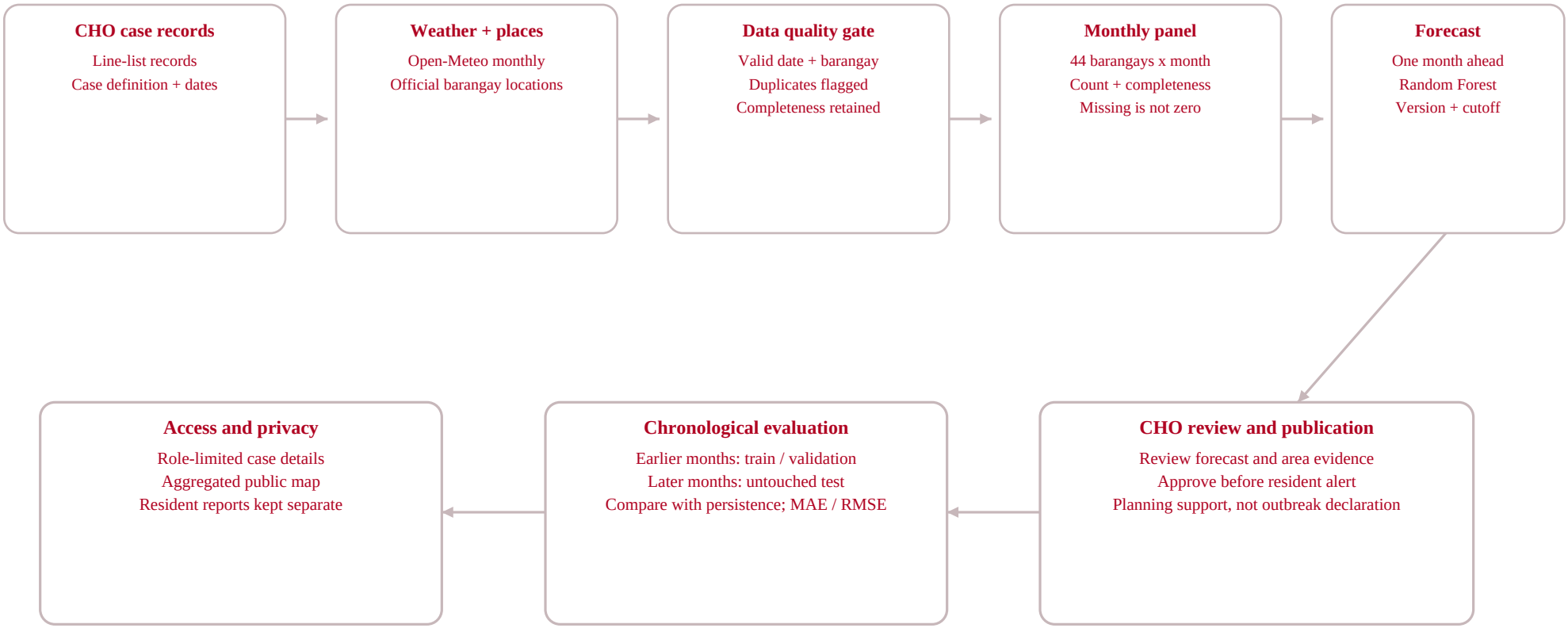
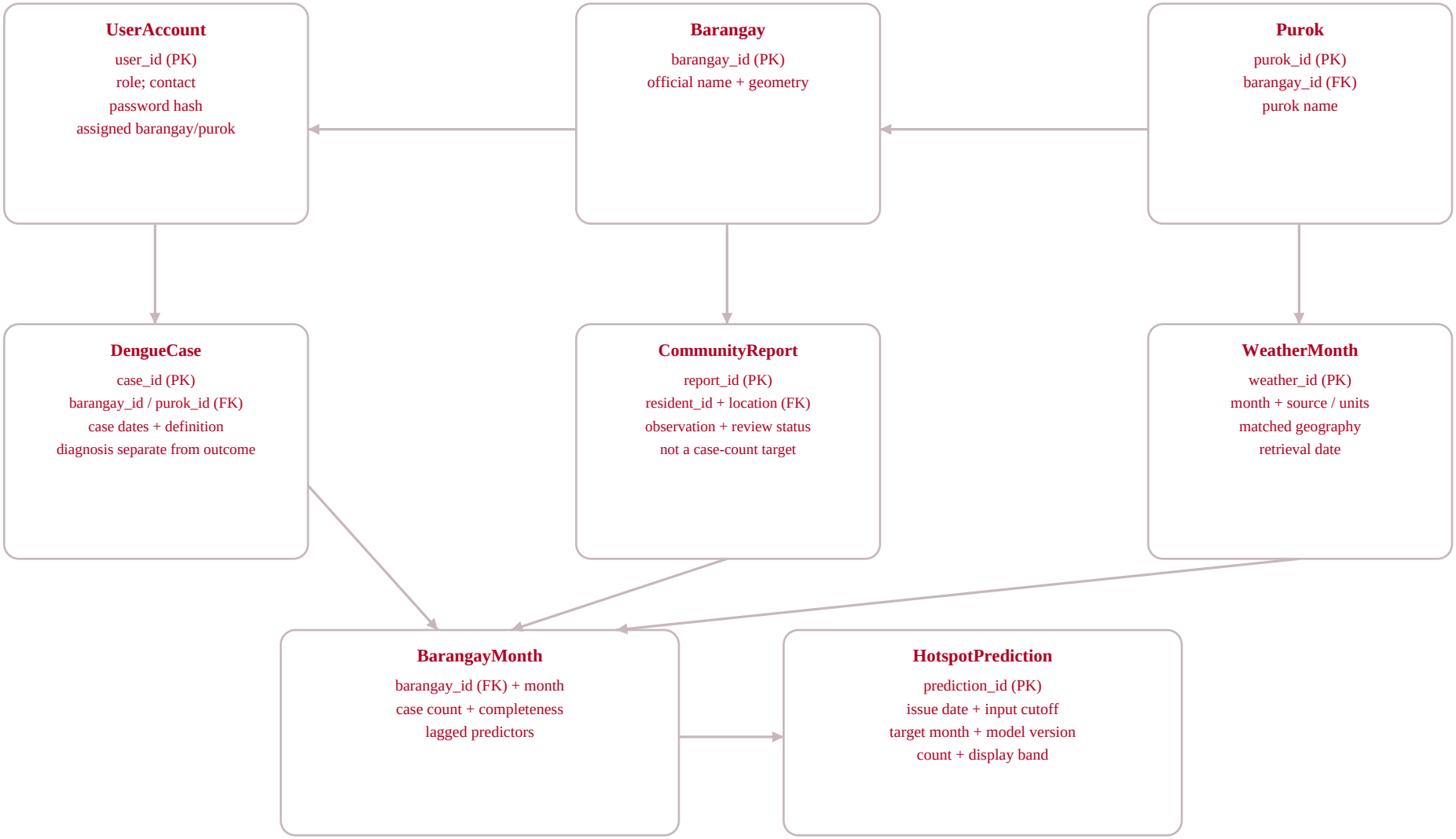


Figure: revised conceptual entity relationships (ERD)



Use integer foreign keys consistently. Keep incomplete reporting unavailable, not zero.